CS351 - IT Workshop II (Cloud Computing)

Assignment 3

Total Marks: 25

Please follow the instructions below. Marks will be deducted if these instructions are not followed.

(i) Screenshots: Take clear screenshots, resize the images to reduce their file size, and upload them to Google Classroom. Please do not upload a .zip file in Google Classroom.

(ii) Source Code File Naming: Name each Python source file using the following format: <assignment_no>_<question_no>.py

For Example: a1_q1.py

(iii) Source Code Header: Include your name and roll number at the top of each source file using the following format.

Name: <Your Name>

Roll No: <Your Roll No>

(iv) Proper Comments: You need to write comments for each block of code, function inputs and outputs, and the working logic of the functions.

(v) Proper Variable and Function Name: Please use meaningful variable and function names.

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1. Create a ***static website*** with one or two HTML pages and copy the pages and dependent files in an ***Amazon S3 bucket***. **[5 Marks]**
2. Create a ***startup script*** to install the ***Apache server*** on an EC2 instance. The script should also copy the website files (from the previous step) from **S3** to the **EC2** instance.

   [3 Marks]
3. Create a Python program using ***boto3*** to launch a ***t3.micro*** instance in EC2. Supply the ***startup script*** you created in the previous step while launching a new instance from this program. **[7 Marks]**
4. Use a ***security group*** with port **80** open. **[2 Marks]**

5. Your program should check the ***status*** of the newly launched instance and return the ***public DNS address*** of the instance when the status changes to ***"running"***. **[6 Marks]**
6. Open the ***public DNS*** of the newly launched instance in a browser and verify if the ***static website*** works. **[2 Marks]**

Please note that you need to install ***boto3*** on your Laptop. You can install ***boto3*** from source or using the ***pip3*** command. For installation and ***boto3*** configuration, you can follow the links below.

https://boto3.amazonaws.com/v1/documentation/api/latest/index.html